Module 3: Cell division, development and disease control

3.1 Cell division and development

This section provides learners with knowledge and understanding of biological development on three levels: cell division by mitosis (where genetic information is copied and passed to daughter cells) cell differentiation, cell division by meiosis and fetal development, and the evolutionary development of species.

Scientific research has highlighted the role of apoptosis during the life cycle of a multicellular organism. The importance of both mitosis and apoptosis is considered in relation to fetal growth and development, as well as an appreciation that these processes are affected by environmental factors. Knowledge of recent advancements in stem cell technology and its potential applications further enhance learners' understanding of these fundamental cellular processes.

Meiosis is also studied as a process that promotes genetic biodiversity and variation. The potential for change and the development of new species by the action of natural selection on genetic variation is seen as an essential part of the process of evolution. Learners will also gain an understanding that the variety of life, both past and present, is extensive, but the biochemical basis of life is similar for all living things as well as of the link between natural selection and the adaptations of living organisms to their environments.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.